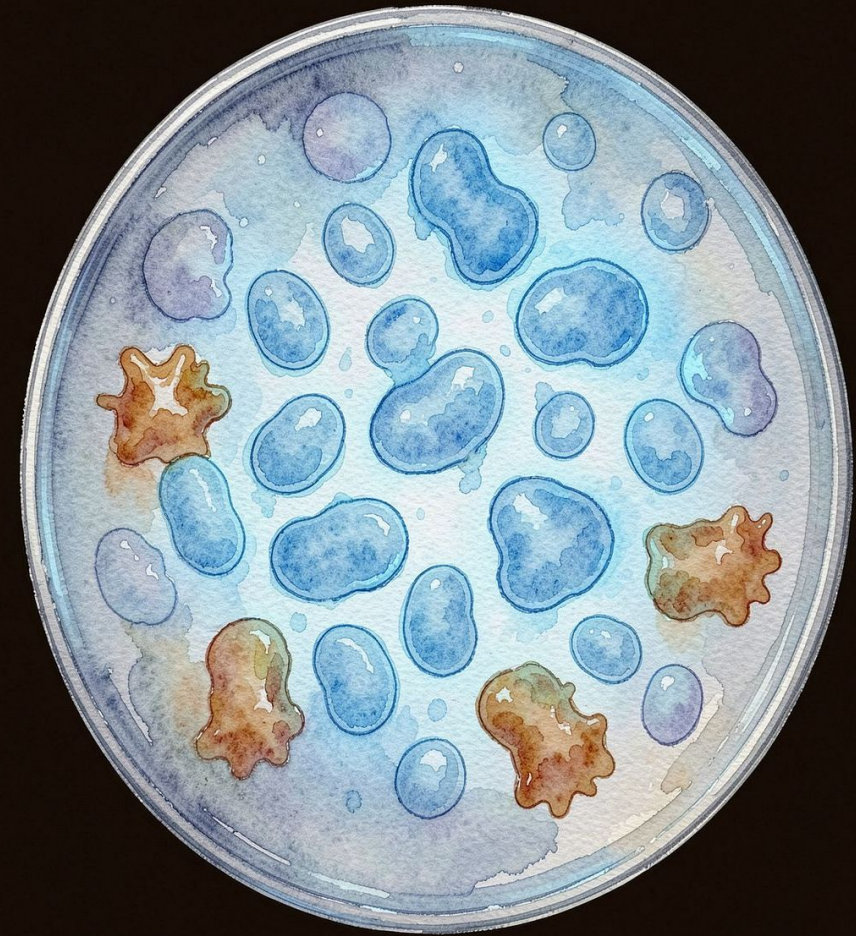


Breast Cancer Classification Using Artificial Neural Networks

A Deep Learning Approach to Medical Image Analysis



The Problem: Early Detection Saves Lives

Global Health Challenge

Breast cancer remains one of the most critical threats to women's health worldwide, demanding urgent and scalable solutions.

Timely Intervention

Early and accurate identification of malignant tumors enables timely medical intervention, significantly improving patient outcomes.

Human Limitations

Manual diagnosis is time-consuming and prone to human error, creating an urgent need for automated classification systems.



Project Objectives

01

Build an ANN Classifier

Design and train an Artificial Neural Network to classify breast tumors as **Benign** or **Malignant**.

03

Effective Preprocessing

Show the impact of feature scaling and preprocessing techniques on model performance and training stability.

02

Achieve High Accuracy

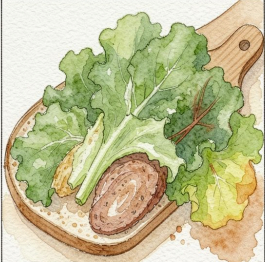
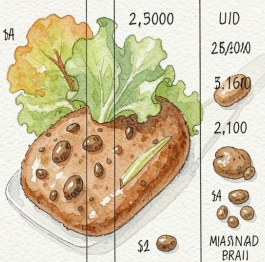
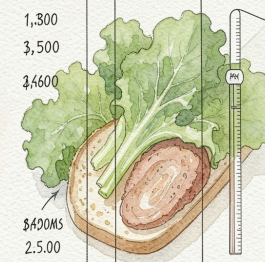
Demonstrate strong generalization by achieving high accuracy on previously unseen test data.

04

Comprehensive Evaluation

Analyze model performance rigorously using a confusion matrix and standard classification metrics.

Dataset Overview

BESTBONC HNEIA HUETNAT		DATA	
		2,5000	UID 25/2010
		1,300 3,500 4,600	5,160 2,100 14 MADNAD BRAH
DAS NAIT # SARAH		\$400MS 2.5.00	

Breast Cancer Dataset – Kaggle

A well-established benchmark dataset for binary tumor classification tasks in machine learning research.

Features include: radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension — each measured across mean, standard error, and worst values.

Target: Binary classification — **Malignant = 1**, Benign = 0

569

Total patient records

30

Numerical features

2

Target classes

Data Preprocessing Pipeline

Separate Features & Target

Isolate X features and y labels for modeling.

Encode Labels

Apply LabelEncoder to the diagnosis column.

Remove ID Column

Eliminate non-predictive identifier to prevent leakage.

Stratified 80/20 Split

Create train/test sets preserving class balance.

Each preprocessing step ensures data integrity and prevents information leakage into the model evaluation process.

Feature Scaling: StandardScaler

Why Scaling Matters for ANNs

Raw medical measurements span vastly different numerical ranges — area values reach thousands while smoothness stays below 1.0. Without normalization, gradient descent converges slowly and erratically.

StandardScaler transforms each feature to **zero mean and unit variance**, placing all 30 features on a comparable scale.

- ❏ The scaler is fitted **only on training data** and then applied to test data — this is essential to prevent data leakage and ensure honest evaluation.

1

Fit on Train

Mean and standard deviation computed exclusively from training samples.

2

Transform Both

Training and test sets scaled using the same training-derived parameters.

3

Faster Convergence

Normalized inputs improve ANN gradient flow and training stability.

ANN Model Architecture

1

Input Layer

30 neurons — one per standardized feature

2

Hidden Layer 1

64 neurons — ReLU activation + 20% Dropout

3

Hidden Layer 2

32 neurons — ReLU activation + 20% Dropout

4

Hidden Layer 3

16 neurons — ReLU activation + 20% Dropout

5

Output Layer

1 neuron — Sigmoid activation

Total Trainable Parameters: 4,609 weights — a compact yet powerful architecture for this classification task.

Model Compilation & Training



Training Configuration

Optimizer: Adam

Adaptive learning rate — combines momentum and RMSProp for fast, stable convergence.

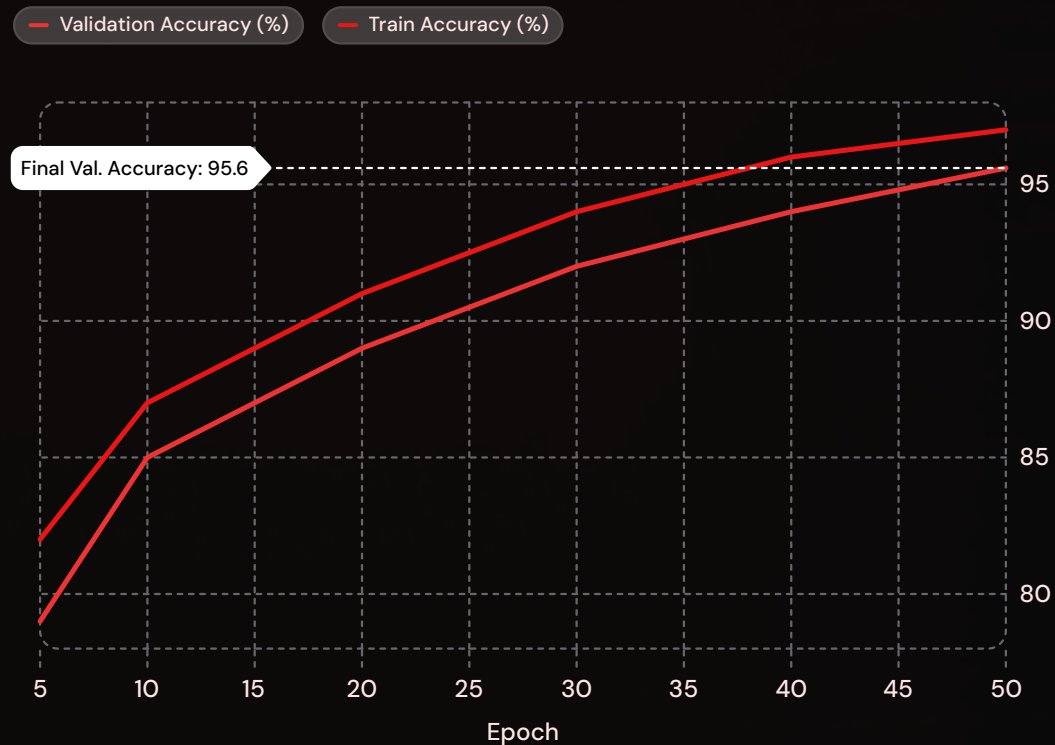
Loss: Binary Crossentropy

The standard loss function for binary classification tasks with sigmoid output.

50 Epochs • Batch Size 16

20% validation split used to monitor generalization throughout training.

Training Results



Steady Learning Progress

The model demonstrated strong and consistent learning across all 50 epochs. Both training and validation accuracy curves converged smoothly, with no signs of overfitting — a direct benefit of the Dropout regularization layers.

- **Final Validation Accuracy:** 95.60%
- **Final Validation Loss:** ~0.137
- Training and validation curves tracked closely, confirming stable generalization throughout the process

Test Performance: **98.25% Accuracy**

98.25%

Test Accuracy

Achieved on 114 unseen test samples

112

Correct Predictions

Out of 114 total test cases

2

Misclassifications

Only 2 incorrect predictions on the entire test set

0.1527

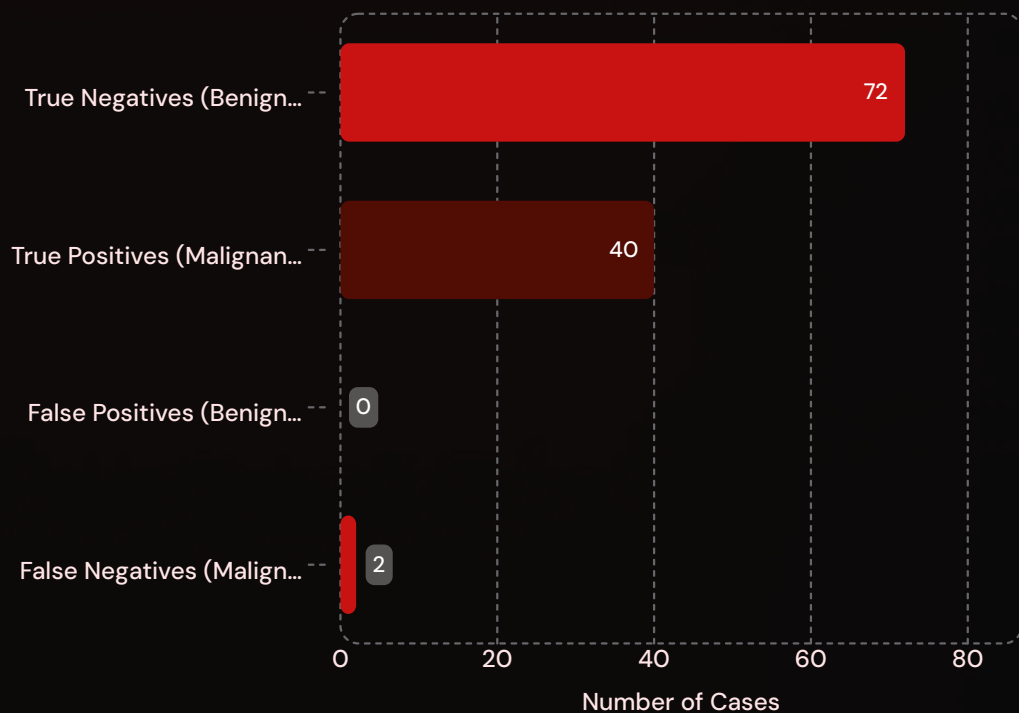
Test Loss

Low loss confirms reliable probability estimates

The model generalizes exceptionally well — surpassing validation accuracy and demonstrating robust performance on previously unseen clinical data.

Confusion Matrix Analysis

Prediction Outcome



Clinical Interpretation

The confusion matrix reveals a clinically significant result: the model produced **zero false positives** — not a single benign case was incorrectly flagged as malignant.

- **72 True Negatives:** All benign cases correctly identified
- **40 True Positives:** Malignant tumors correctly caught
- **0 False Positives:** No unnecessary alarms raised
- **2 False Negatives:** Two malignant cases missed — the primary area for future improvement

❏ Minimizing false negatives remains the key optimization goal for any clinical deployment of this model.